www.tntextbooks.in

5.13.3 Family: Solanaceae (Potato Family / Night shade family)

Systematic Position

APG system of classification		Bentham and Hooker system of classification	
Kingdom	Plantae	Kingdom	Plantae
Clade	Angiosperms	Class	Dicotyledonae
Clade	Eudicot	Subclass	Gamopetalae
Clade	Asterids	Series	Bicarpellatae
Clade	Solanales	Order	Polemoniales
Family	Solanaceae	Family	Solanaceae



Diagnostic Features

- Leaves alternate, exstipulate.
- Flowers actinomorphic, pentamerous.
- Calyx often persistence / accrescent.
- Stamens 5, epipetalous, poricidal in dehiscence.
- Carpels 2, ovary superior, 2 chambered, obliquely placed, falsely four chambered placenta swollen, ovule numerous.
- Fruits berry or capsule, vascular bundles with both outer and inner phloem (Bicollateral vascular bundle).

General Characters

Distribution:

Family Solanaceae includes about 88 genera and about 2650 species, of these *Solanum* is the largest genus of the family with about 1500 species. Plants are worldwide in distribution but more abundant in South America.

Habit: Mostly annual herbs, shrubs, small trees (*Solanum violaceum*) lianas with prickles (*Solanum trilobatum*), many with stellate trichomes; rarely vines (*Lycium sinensis*)

Root: Branched tap root system.

Stem: Herbaceous or woody; erect or twining, or creeping; sometimes modified into tubers (*Solanum tuberosum*) often with **bicollateral** vascular bundles.

Leaves: Alternate, simple, rarely pinnately compound(*Solanum tuberosum* and *(Lycopersicon esculentum*) exstipulate, opposite or sub-opposite in upper part, unicostate reticulate venation.

Inflorescence: Generally axillary or terminal cymose (*Solanum*) or solitary flowers (*Datura stramonium*). Extra axillary scorpiod cyme called **rhiphidium** (*Solanum nigrum*) solitary and axillary (*Datura* and *Nicotiana*) umbellate cyme (*Withania somnifera*).

Flowers: Bracteate (*Petunia*), or ebracteate (*Withania*) pedicellate, bisexual, heterochlamydeous, actinomorphic or weakly zygomorphic due to oblique position of ovary pentamerous, hypogynous.

Calyx: Sepals 5, rarely 4 or 6, Synsepalous, valvale peristaent, often accrescent (enlarging to envelop the fruit) occasionally enclosing the fruit (*Physalis, Withania*)

Corolla: Petals 5, sympetalous, rotate, tubular (*Solanum*) or bell- shaped (*Atropa*)